

e.g. $\frac{dy}{dx} = 5y$

$$\int \frac{1}{y} dy = \int 5 dx$$

$$\ln|y| = 5x + c$$

$$y = e^{5x+c}$$

$$y = e^{5x} e^c$$

$$y = k e^{5x}$$

8. Separable Equations: The most fundamental technique approach. You algebraically isolate y (or x) on one side and x (or y) on the other.

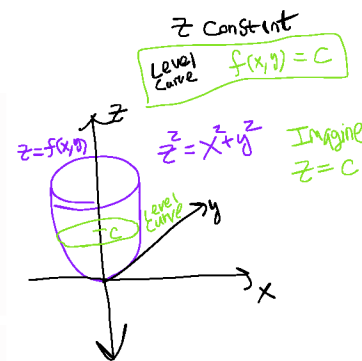
- Equation $\frac{dy}{dx} = g(x)h(y)$
- Process: Rearrange to $\frac{1}{h(y)} dy = g(x) dx$ and integrate both sides.
- Linear ODEs: Integrating factors (but are variables cannot be easily separated).
- Standard Form: $\frac{dy}{dx} + P(x)y = Q(x)$
- Process: Compute the integrating factor $\mu(x) = e^{\int P(x) dx}$. Multiply the entire differential equation by $\mu(x)$. The linear form now allows the exact derivative of a function $\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$. Integrate both sides to solve for y .
- Exact Equations: Based on the relationship between M and N .
- Form: $M(x,y)dx + N(x,y)dy = 0$
- Condition: It is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
- Process: Integrate M with respect to x for V with respect to y to find the potential function $f(x,y) = C$.
- Approximating Solutions: When analytical solutions fail, numerical approximations step in.
 - Euler's Method: Formulas $y_{n+1} = y_n + h \cdot f(x_n, y_n)$
 - Process: Using an initial value (x_0, y_0) and a step size h , you iteratively calculate the next point by following the tangent line. The algorithmic approach is excellent for coding dynamic models.
- Homogeneous Linear 2nd-Order Equations: The foundation for finding the "complementary" solution y_c .
 - Equation: $ay'' + by' + cy = 0$
 - Process: Assume a solution of the form $y = e^{rt}$. This yields the characteristic equation $ar^2 + br + c = 0$. The roots of this algebraic quadratic (irreducible, repeated, or complex) dictate the form of the solution.
- Method of Undetermined Coefficients: Used for non-homogeneous equations $ay'' + by' + cy = g(x)$ where $g(x)$ is basic (polynomial, exponential, or sinusoidal).
 - Process: Formulate an educated guess for the particular solution y_p based on the "form" of derivatives of $g(x)$. Plug the guess into the ODE and solve for the unknown coefficients.
- Variation of Parameters: A more robust method for finding y_p when Undetermined Coefficients fails.
 - Process: Uses the complementary solutions y_1 and y_2 .
 - Equations: $y_p = -y_1 \int \frac{y_2 g(x)}{W} dx + y_2 \int \frac{y_1 g(x)}{W} dx$, where W is the Wronskian determinant $W(y_1, y_2) = y_1 y_2' - y_1' y_2$.
- Mass-Spring Oscillator: The primary physical application of 2nd-order ODEs.
 - Equation: $m\ddot{x} + b\dot{x} + kx = F(t)$ (where m is mass, b is the damping constant, and k is the spring constant).
- Phase Diagrams and the Phase Plane: A geometric method to analyze the stability of solutions without explicitly solving the equation.
 - The Asymptote Method: Finding a differential operator L that "annihilates" $F(t)$ to avoid the tedious process of $g(x)$, effectively turning a non-homogeneous problem into a higher-order homogeneous one.

The Laplace Transform is a paradigm shift. Instead of solving with calculus in the time domain (t) , it converts the entire differential equation into an algebraic equation in the frequency domain (s) .

- Definition of the Laplace Transform:
 - Equation: $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$
- Solving ODEs with Laplace:
 - Process: Apply the transform to both sides of the ODE. Use the derivative property $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$ to translate initial conditions automatically. Algebraically solve $F(s)$, then apply the inverse Laplace Transform, $\mathcal{L}^{-1}$, to return to the time domain $f(t)$.
- Transforms of Discontinuous Functions: Functions using the Heaviside Step Function $u(t-a)$ which acts as an "on/off" switch for functions of time.
 - Equation: $\mathcal{L}\{u(t-a)\} = e^{-as}$
- Convolution: A method to find the inverse transform of a product of two s -domain functions.
 - Equation: $\mathcal{L}^{-1}\{F(s)G(s)\} = \int_0^t f(\tau)g(t-\tau) d\tau$

When functions have variable coefficients (like $x^2 y'' + xy' + y = 0$) and standard analytic methods fail, we require the solutions as infinite power series.

- The Taylor Polynomial & Power Series:
 - Assumption: Assume the solution takes the form $y(x) = \sum_{n=0}^\infty a_n (x-x_0)^n$.
 - Process for Power Series Solutions:
 - Differentiate the series to find y' and y'' .
 - Substitute these series back into the original differential equation.
 - Shift the summation indices to align all terms to the same power of x .
 - Factor out the a_n terms to establish a recurrence relation: an algebraic rule that defines each coefficient a_n in terms of previous coefficients.
 - Analytic Functions and Existence: Analyzing the differential equation to determine "ordinary points" (where power series solutions are guaranteed to behave well) versus "singular points" (where the series might break down or require the Frobenius method).



$f(x,y) = C$

$z = \text{constant}$

Level Curve $f(x,y) = C$

$z = x^2 + y^2$

Imagine $z = C$

$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$

$\frac{d}{dx} f(x,y) = \left[\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} \right] = \frac{dC}{dx} = 0$

Exact Diff. Eq.

Consider a population model involving two interacting species x and y constrained by the exact differential equation $2xyz dx + (x^2 + 3y^2) dy = 0$. What is the implicit solution representing the conserved relationship $F(x,y) = C$?

$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$

Exact Differential Equation

$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$

$f(x,y) = C \iff M dx + N dy = 0$

$M = \frac{\partial f}{\partial x} = f_x$

$N = \frac{\partial f}{\partial y} = f_y$

$M_y = \frac{\partial f}{\partial x \partial y} = f_{xy} = N_x = \frac{\partial f}{\partial y \partial x} = f_{yx}$

If $M_y = N_x$, then f exists & $f(x,y) = C$

Recall $f_{xy} = f_{yx}$

$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$

$f = \sin x + ye^x + x^2 y^2$

$f_x = \cos x + ye^x + 2xy^2$

$f_y = e^x + 2x^2 y$

$f_{xy} = e^x + 4xy$

$f_{yx} = e^x + 4xy$

$f_{xx} = -\sin x + e^x + 2y^2$

$f_{yy} = 2x^2$

$f_{xy} = e^x + 4xy$

$f_{yx} = e^x + 4xy$

$f_{xx} = -\sin x + e^x + 2y^2$

$f_{yy} = 2x^2$

$f_{xy} = e^x + 4xy$

$f_{yx} = e^x + 4xy$

3. The Green Box: "Imagine $z=C$ "

This is where Exact Differential Equations are born!

If $f(x,y) = C$ (where C is a constant number like 500 feet), it means you are walking on a Level Curve—a perfectly flat path wrapping around the mountain.

Because your altitude is constant, your total change in altitude (df) is exactly zero.

If you take the two-part recipe from above and set the total change to zero, you get the fundamental formula for an exact differential equation:

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

(Which is usually written as $M dx + N dy = 0$)

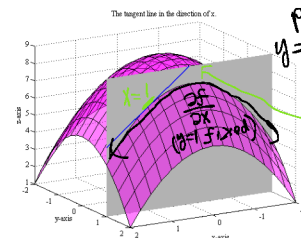
4. The Bottom Text: $\frac{d}{dx} f(x,y) = \frac{\partial f}{\partial x}$

This is the math tying directly back to your first image.

What happens if you don't walk diagonally? What if you freeze your sideways movement entirely, just like the gray slicing plane did in the last image?

If you don't move sideways, your sideways step ($\frac{dy}{dx}$) becomes 0. That completely wipes out the entire second half of the recipe.

You are left with only the first part: the total change ($\frac{d}{dx}$) is exactly equal to the partial change ($\frac{\partial f}{\partial x}$).



Plane $y=1$

$\frac{\partial f}{\partial x}$

$\frac{\partial f}{\partial x}(1)$

$\frac{\partial f}{\partial x}(1)$

$\frac{\partial f}{\partial x}(1)$

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1. The Mountain (The 3D Shape)

Imagine the purple grid shape is a physical hill. The mathematical function $f(x,y)$ is just a machine that tells you your altitude (the z -axis) based on your exact GPS coordinates (your x and y positions).

In standard 2D calculus, you only had an x -axis, so finding the "steepness" (the slope) was easy: you just measured how steep the line was. But on a 3D hill, you can walk North, South, East, West, or anywhere in between. The steepness changes depending on which way you face.

2. The Slice (The Gray Plane)

Because walking in any random direction is too complicated, we force a rule: we are only allowed to walk in one specific direction. "y is fixed!" Look at the gray wall slicing through the mountain. This represents freezing your position on the y -axis at exactly 1.

By freezing y , you are essentially taking a giant meat cleaver, chopping the 3D mountain in half, and throwing the rest away.

You are now trapped on that flat, 2D gray wall. You can only move back and forth along the x -axis.

3. The Blue Line (The Partial Derivative)

Now that you are trapped on that 2D slice, the problem is exactly the same as the basic algebra you learned in high school.

The blue line is the tangent line. It is a straight line resting on the curve that shows you exactly how steep the hill is at that specific green spot ($x = 1$).

The green text "slope of this tangent line" is the final answer. It is just a number telling you how fast you are going up or down if you take a step in the x -direction.

The Math Translation

The purple text at the top left is explaining the notation:

$\frac{d}{dx} f(x,y)$. This is the old-school way of writing a derivative.

$\frac{\partial f}{\partial x}$: this is the multivariable way of writing a derivative. The curly ∂ (called "partial") is just a warning label. It tells you (and is reading the math: "hey, this hill actually has other directions you can walk like y), but we are treating them as dead, frozen numbers right now to find the slope in the x -direction."

In the simplest possible terms: A partial derivative is just finding the steepness of a hill while forcing yourself to only walk in one straight compass direction.

Product Rule

origins $\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + u'v = \frac{Q(x)}{u}$

$\rightarrow \frac{dy}{dx} + \frac{u'}{u}y = \frac{Q(x)}{u}$ 1st order linear Diff. EQ

Linear ODEs & Integrating Factors (used when variables cannot be easily separated)

- Standard Form: $\frac{dy}{dx} + P(x)y = Q(x)$
- Procedure: Compute the integrating factor $\mu(x) = e^{\int P(x)dx}$. Multiply the entire differential equation by $\mu(x)$. This forces the left side to become the exact derivative of a product: $\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$. Integrate both sides to solve for y .

Standard form $\frac{dy}{dx} + 3x^2y = e^x$

Solve $\mu = e^{\int 3x^2 dx} = e^{x^3} = e^{x^3}$

So $\mu = e^{x^3}$

Sol'n $y = \frac{1}{\mu} \left[\int \mu Q dx + C \right]$

$= \frac{1}{e^{x^3}} \left[\int e^x e^{x^3} dx + C \right]$

$y = \frac{1}{e^{x^3}} \left[\int e^{x^3+x} dx + C \right]$

Standard Form: $\frac{dy}{dx} + P(x)y = Q(x)$

Solve $x \frac{dy}{dx} + 2y = 4x^2$, $y(1) = 2$.

find $y(x)$

Standard form

$\frac{dy}{dx} + \frac{2}{x}y = 4x$

$\mu = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln(x^2)} = x^2$

$\mu = x^2$

$y = \frac{1}{\mu} \left[\int \mu Q dx + C \right]$

$(uv)' = u'v + uv'$

$\mu \left[\frac{dy}{dx} + \frac{2}{x}y = 4x \right]$

$= x^2 \frac{dy}{dx} + 2xy = 4x^3$

$= \int (x^2 y)' dx = \int 4x^3 dx$

$\left[2xy + x^2 \frac{dy}{dx} \right]$

$Q = 4x$

$\mu Q = 4x^3$

$\int \mu Q dx = \int 4x^3 dx = x^4 + C$

$\int (x^2 \cdot 4x) dx = \int (4x^3) dx$

$x^2 y = \frac{x^4 + C}{x^2}$

$y = \frac{1}{x^2} [x^4 + C]$

$= \frac{1}{\mu} [\mu Q dx + C]$

$\mu = x^2$

$y = \frac{1}{x^2} \left[\int x^2 \cdot 4x dx + C \right]$

$= \frac{1}{x^2} [x^4 + C]$

$y = \frac{x^4 + C}{x^2}$

$\mu \frac{dy}{dx} + \mu P(x)y = \mu Q(x)$

Force $\frac{d\mu}{dx} = \mu P(x) \Leftrightarrow \int \frac{1}{\mu} d\mu = \int P(x) dx$

$\frac{d\mu}{dx} = \mu P(x) \Leftrightarrow \int \frac{1}{\mu} d\mu = \int P(x) dx$

$\ln|\mu| = \int P(x) dx$

$\mu = e^{\int P(x) dx}$

$\mu y' + \mu' y = \mu Q(x)$

$\int (\mu y)' dx = \int \mu Q(x) dx$

$\mu y = \frac{\int \mu Q dx + C}{\mu}$

$y = \frac{1}{\mu} \left[\int \mu Q dx + C \right]$

$$(f(x,y) = C)$$

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

$$M_y = \frac{\partial f}{\partial x \partial y} = f_{xy}$$

$$M dx + N dy = 0$$

$$N = \frac{\partial f}{\partial y}$$

$$M = \frac{\partial f}{\partial x}$$

Consider a population model involving two interacting species x and y constrained by the exact differential equation $2xy dx + (x^2 + 3y^2) dy = 0$. What is the implicit solution representing the conserved relationship $F(x,y) = C$?

$$\text{Test } M_y \stackrel{?}{=} N_x$$

$$M = 2xy$$

$$M_y = 2x$$

$$(f_{xy}) (f_{yx})$$

$$N = x^2 + 3y^2$$

$$N_x = 2x$$

$$M_y = N_x \text{ Exact!}$$

goal:
solve for
 $f(x,y) = C$

$$M = f_x$$

$$\rightarrow f = \int M dx$$

partial
integral
w.r.t. x

$$f = \int (2xy) dx$$

$$= 2y \int x dx$$

$$= 2y (\frac{1}{2} x^2) + C(y)$$

$$f = yx^2 + C(y)$$

$$\left[f_x = 2yx + 0 \right]$$

$$N = f_y$$

$$f = \int N dy$$

partial
integral
w.r.t. y

$$f = \int (x^2 + 3y^2) dy$$

$$f = x^2 y + y^3 + C(x)$$

$$\underline{yx^2} + \underbrace{C(y)}_{y^3} = \underline{x^2 y} + \underbrace{y^3}_{C(y)} + \cancel{C(x)}$$

$$f(x,y) = yx^2 + y^3$$

$$\text{Solution } f(x,y) = C$$

$$yx^2 + y^3 = C$$

Multivariable
Chain Rule

$$f(x,y) = C \rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

$$= M dx + N dy = 0$$

$$M = \frac{\partial f}{\partial x}, \quad N = \frac{\partial f}{\partial y}$$

$$\rightarrow f = \int M dx, \quad f = \int N dy$$

7. To solve the exact equation $(\cos x - y \sin x)dx + (\cos x)dy = 0$, what is the level curve equation $F(x, y) = C$?

① Exact?
Is $M_y = N_x$?
yes

$$M = \cos x - y \sin x$$

$$N = \cos x$$

$$M_y = -\sin x$$

$$N_x = -\sin x$$

② find f

$$f = \int M dx = \int (\cos x - y \sin x) dx$$

$$f = \int N dy = \int \cos x dy$$

$$= \underbrace{\sin x}_{C(x)} + \underbrace{y \cos x}_{C(y)} = y \cos x + C(x)$$

$$f(x,y) = \sin x + y \cos x$$

orig. level
curve

$$(f(x,y) = C)$$

$$\sin x + y \cos x = C$$

Ordinary Differential Equations

Exam 1 — Comprehensive Study Guide

Chapters 1–3 • First-Order Methods, Qualitative Analysis & Second-Order Linear Theory

How to use this guide. Work through it in four passes: (1) lock in the vocabulary so you can classify any equation on sight; (2) memorize the algorithms & formula sheet; (3) re-derive the four worked masters from a blank page until they are automatic; (4) attempt the 8 practice problems cold, then grade yourself against the detailed key. Every algebraic transition below is written out in full — no skipped steps.

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1. Core Vocabulary & Concept Glossary

Order of a differential equation
The order is the highest derivative that appears. $y' + 3y = t$ is first order; $y'' + 4y' + 13y = 0$ is second order; $y^{(4)} = 0$ is fourth order.
Linear vs. Non-linear
An n th-order ODE is linear if it can be written $a_n(t)y^{(n)} + \dots + a_1(t)y' + a_0(t)y = g(t),$ i.e. the unknown y and all of its derivatives appear only to the first power, never multiplied together, and never inside a nonlinear function. The coefficients $a_k(t)$ and the forcing $g(t)$ may depend on t in any way. An equation is non-linear if a derivative appears as y^2 , yy' , $\sin y$, e^y , $\sqrt{y'}$, etc.

Linear 2nd Order Differential Equations

Complete Worksheet with Detailed Solutions

Question 1. Score: 1/1

Determine the order of each differential equation.

- $d^4y/dt^4 = \sin(t + y) \rightarrow$ Order: **4**
- $yy' + t^2y = t^4 \rightarrow$ Order: **1**
- $d^4y/dt^4 - t \cdot d^2y/dt^2 + 5y = 1 \rightarrow$ Order: **4**
- $5y' - 2y^4 = 0 \rightarrow$ Order: **1**
- $t^2y'' + t^4y' + 5y = \sin(t) \rightarrow$ Order: **2**
- $\cos(t) \cdot d^2y/dt^2 = 5y \rightarrow$ Order: **2**

Solution:

The **order** of a differential equation is determined by the **highest derivative** present.

- $d^4y/dt^4 = \sin(t+y)$: highest derivative is 4th $\rightarrow$ **Order 4**
- $yy' + t^2y = t^4$: highest derivative is y' (1st) $\rightarrow$ **Order 1**
- $d^4y/dt^4 - t \cdot d^2y/dt^2 + 5y = 1$: highest is 4th $\rightarrow$ **Order 4**
- $5y' - 2y^4 = 0$: y^4 is y to the power 4, NOT a 4th derivative; highest derivative is $y' \rightarrow$ **Order 1**
- $t^2y'' + t^4y' + 5y = \sin(t)$: highest derivative is y'' (2nd) $\rightarrow$ **Order 2**
- $\cos(t) \cdot y'' = 5y$: highest derivative is $y'' \rightarrow$ **Order 2**

■ **Key Concept:** The order of an ODE is the order of the highest derivative appearing in the equation. Exponents on y itself (like y^4) do not affect the order.

Question 5. Score: 0/3

Solve the following initial value problem.

$y' + 2x(y + 1) = 0, y(0) = 2$

Student answer (incorrect): $y(x) = 3e^{-x^2} - 1$

Solution:

This is a **separable** first-order ODE. Rewrite:

$y' = -2x(y + 1)$

Separate variables (divide both sides by $(y+1)$, assuming $y \neq -1$):

$dy/(y+1) = -2x \, dx$

Integrate both sides:

$\ln|y + 1| = -x^2 + C$

Exponentiate:

$y + 1 = Ae^{-x^2}$ where $A = \pm e^C$

Apply initial condition $y(0) = 2$: